SELECTED CORROSION PROPERTIES OF A NEW AMORPHOUS AL-CO-CE ALLOY SYSTEM

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ABSTRACT

Selected corrosion properties of the recently developed amorphous Al-Co-Ce alloy system are reported. Improved resistance to halide-induced pitting was observed in near neutral solution when the fully amorphous alloy and an aged alloy (100°C for up to1000 hrs.) were compared to high purity aluminum. Both the open circuit potential and the repassivation potential were found to be strongly influenced by the concentration of cobalt in the alloy. Tunable corrosion properties based on pre-selection of Co content is shown to be feasible. This alloy system also showed improved resistance to rapid dissolution in alkaline solutions, in contrast with the behavior of pure aluminum. However, in acidic solutions the dissolution rate of the amorphous alloy was increased when compared to pure aluminum. These unique properties of the amorphous alloys remained after structural relaxation and/or partial devitrification by low temperature heat-treatments at 100°C. However, complete recrystallization by 550°C heat treatment removed all benefits seen in amorphous variants.

Keywords: amorphous alloys, local corrosion, passivity, devitrification, Al-Co-Ce alloys, tunable corrosion properties, depassivation

INTRODUCTION

Over the past several years, a wide range of novel amorphous alloys have been produced with ETM-LTM-SM compositions (where ETM-LTM-SM signifies Early Transition Metal-Late Transition Metal-Simple Metal). These amorphous alloys offer significant enhancements of both mechanical strength and corrosion resistance over their fully crystalline counterparts[1-5]. One subset of the ETM-LTM-SM compositions showing particular promise as high-strength low-density material is the class of Al-TM-RE alloy compositions. These alloys have demonstrated high glass-forming abilities as well as improved corrosion resistance over conventional aluminum alloys[6,7]. In addition this class of alloys possesses the capacity to develop nanocrystals in an amorphous matrix upon appropriate heat treatment[8]. Such nanocrystals have been shown to further enhance the mechanical properties without adversely affecting the improved corrosion resistance normally associated with fully amorphous alloys[9,10].

One alloy system within the Al-TM-RE class of amorphous alloys that has recently been characterized is Al-Co-Ce [11,12]. This system has shown to have high thermal stability, as well as a large compositional range over which an amorphous microstructure can be obtained[13]. By making use of these unique physical characteristics, as well as the beneficial influence which transition metals have on the barrier corrosion properties[7] and sacrificial coating capabilities [3,5] when added to solid solution of amorphous aluminum, the opportunity exists to create a multifunctional metallic coating. Additional benefit of this system as a coating comes from the possibility of an active corrosion inhibition function provided by the release and delivery of the oxidation products of Co and Ce. Although no known previous study has focused on the release of these elements from solid solution of aluminum, Co(II) and Ce (III) have shown great promise as inhibitors when deposited on the surface of aluminum from aqueous solution[13-19]. In addition, recent preliminary studies have demonstrated active corrosion inhibition where an Al-Co-Ce coating is placed in close proximity to 2024-T3 through release and transport of Co(II) and Ce(III) in the aqueous solution[20].

In order to optimize each of these coating functions, a full understanding is needed of each alloying element's individual influence on corrosion properties. By examining the corrosion behavior over a wide compositional range of amorphous alloys within the Al-Co-Ce system, the effect of both Co and Ce can be isolated, resulting in a tunable alloy system. This study reports on selected corrosion properties as they relate to alloy composition within the amorphous Al-Co-Ce system. Specifically, dissolution over a range of pH values is examined, resistance to pitting in halide solutions is explored, and lastly the examination of the open circuit potentials is carried out to investigate the possibility that this alloy could function as a sacrificial anode.

EXPERIMENTAL

To assess the barrier corrosion properties of various Al-Co-Ce alloy compositions, a set of diagnostic parameters is needed to quantify each composition's corrosion behavior. Included in the properties to be assessed were the pitting resistance in halide solution, the dissolution rate at various pH values, and the mixed potential behavior over a range of pH values. The four parameters chosen to characterize these corrosion properties in this preliminary report were the open-circuit potential (OCP), passive current density (i_{pass}), pitting potential (E_{pit}), and repassivation potential (E_{rp}). The pitting and repassivation potentials served as indicators of the pitting resistance, while the passive current density was used to assess the dissolution rate as a function of pH. Open-circuit potentials were gathered to assist in gauging the galvanic corrosion behavior of this alloy (e.g., potential driving force for sacrificial cathodic protection when coupled to AA2024-T3).

Al-Co-Ce amorphous metal ribbons were prepared by using a single-wheel melt-spinning technique under a helium atmosphere. A circumferential wheel speed of 52 m/s was used to produce lengths of ribbon that were approximately 20 μm thick and 2 mm wide. The ingots used in producing these ribbons were prepared from materials with the following purities: >99.999% Al, >99.995% Co, and >99.9% Ce. The compositions of the alloys examined ranged from 0-10 atomic % cobalt and 3-11 atomic % cerium with aluminum composing the remainder (Table I). These particular compositions were selected for analysis due to their either wholly amorphous or largely amorphous + isolated crystalline microstructures, as shown in the Al-Co-Ce structural map developed by Unlu and Inoue[21] for alloy ribbons produced with a wheel speed of 52 m/s (Figure 1). Following their production, non-crystalline microstructures were verified through x-ray diffraction (Figure 2). The diffraction measurements were conducted with a Scintag X-Ray diffractometer using Cu K_α radiation at 40 kV and 35 mA settings in the 2θ range from 10° to 80°. While it is well known that x-ray diffraction (XRD) cannot differentiate between wholly amorphous microstructures and those containing small volume fractions of nanocrystals in an amorphous matrix[22], it has also been shown that most of the parameters examined in this study are unaffected by the presence of nanocrystals less than 10 nm in diameter[9, 23]. However, one parameter that has been shown to be influenced by the presence of any structural heterogeneity in otherwise amorphous materials, such as nanocrystals, is E_{pit}[9, 10]. Therefore, conclusions cannot be made on the compositional dependence of E_{pit} within the Al-Co-Ce alloy system without a thorough characterization of the nano-scale microstructure as a function of composition.

To examine the retention of barrier properties for this alloy system following low temperature heat treatments, a series of artificial aging treatments were carried out for the $Al_{87}Co_7Ce_6$ composition. These treatments result in either devitrification of the initially amorphous material via the formation of isolated small-scale crystalline regions or relaxation of the as-spun amorphous state. In an approach similar to investigations previously conducted on the Al-Fe-Gd system[23], the amorphous ribbon was placed in a furnace maintained at 100°C, and exposed for 1, 10, 100, and 1000 hours. In addition, a high-temperature 24-hour exposure at 550°C was also carried out to induce complete recrystallization. No encapsulation was necessary due to the polishing procedures used to remove the effects of thermal oxidation. Complete microstructural characterization was conducted by Csontos[24] on similarly treated Al-Fe-Gd amorphous alloys to reveal isolated nanocrystal development at the lower temperature treatments and full recrystallization occurring when samples were treated at 550°C. To confirm a similar response to heat-treatments for the Al-Co-Ce system, XRD was conducted on 24-hour 550^{o}C exposed and 100-hour 100^{o}C exposed $Al_{87}Co_7Ce_6$ material. As can be seen from Figure 2 and ?, the high-temperature treatment resulted in full recrystallization and the low-temperature treatment produced aluminum nanocrystals.

Ribbon samples were prepared for electrochemical experiments by mounting on end in Buehler® low-viscosity epoxy resin which allowed the cross-section of the ribbon (area ~10^{-4} cm^2) to be exposed when testing in a conventional electrochemical flat cell. This resin had a curing temperature of 28°C, which maintained the samples near room temperature during mounting to prevent inadvertent nanocrystal formation due to thermal effects[9]. The sample surface was prepared before each test by wet-polishing on SiC paper to a grit of 1200, followed by a final polish with 1 μm alumina suspension and then rinsing with deionized water.

Electrochemical tests were carried out using a PAR 273A potentiostat under control of CorrWare™ corrosion measurement software. In addition to the working electrode, the three-electrode cell was equipped with a platinum mesh counter electrode, and a saturated calomel reference electrode. Potential values are reported versus SCE for all tests conducted in this study. The solutions used were prepared with deionized water and ACS certified reagents, and included 0.6 M NaCl solution adjusted to different pH values using either 1 M NaOH or 1 M HCl.

The typical experimental test began with a one hour open-circuit exposure to 25°C nitrogen-purged solution, immediately followed by a potentiodynamic scan starting at –0.1 V_{ocp}, then anodically polarizing at a rate of 0.1 mV/sec until reversing scan direction when current density reached a threshold of 1,000 $\mu A/cm^2$. From these scans the four diagnostic parameters were extracted, including OCP, i_{pass}, E_{pit}, and E_{rp}. When the E_{pit} and E_{rp} parameter values from potentiodynamic scans were not clear, E_{pit} was taken as the potential at which the current density reached a value two orders of magnitude above i_{pass} and E_{rp} was taken as the potential where the current density decreased below 10^{-2} A/cm^2 on the reverse scan. Although the pit depth and total charge were comparable from trial to trial there was some minor fluctuation in these values. However, examination of the results showed E_{rp} values to be independent of the small variances in either pit depth, total charge, or exposed electrode area.

For each composition, multiple scans (10-20) were carried out to obtain a statistically accurate value for each of the parameters. This data was then used to construct cumulative probability plots, which compared the parameter values for all alloy compositions tested. Mean values from each composition were also calculated and used to plot the corrosion parameter values as a function of each alloying element. This revealed the influence of the individual elements in controlling the corrosion behaviors. As mentioned previously, the relationship between E_{pit} and specific elemental composition within the Al-Co-Ce system could not be accurately developed due to the possible influence of nano-scale heterogeneities undetectable with standard XRD techniques.

Finally, additional scans were conducted on similarly mounted high-purity aluminum (99.999%) and Al 2024-T3 for comparison purposes. The results from these tests were plotted along with those obtained from the Al-Co-Ce system to gauge the differences in corrosion behavior between the amorphous alloys and the conventional materials.

RESULTS

PH Dependent Corrosion Behavior

The influence of pH on both OCP and dissolution rate was examined for a variety of the alloy compositions within the Al-Co-Ce system, and the results were compared with similar tests conducted on AA 2024-T3 (Figure 3). Both the AA 2024-T3 and the amorphous alloys in the Al-Co-Ce system exhibited a decrease in the OCP as the pH values increased from 2 to 12 in deaerated 0.6 M NaCl solution. When AA 2024-T3 was tested in deaerated solution, the OCP ranged from –1.109 V_{SCE} in pH 4 solution to –1.224 V_{SCE} in pH 12 solution. Although the Al-Co-Ce behavior varied depending on the particular alloy composition, each alloy showed a decrease in OCP as the solution pH was raised. When the pH increased from 3 to 10, the OCP of $Al_{90}Co_3Ce_7$ decreased by 0.225 V_{SCE}, $Al_{84}Co_5Ce_{11}$ decreased by 0.121 V_{SCE}, and $Al_{87}Co_7Ce_6$ decreased by 0.150 V_{SCE}. Moreover, the OCP at a given pH decreases with decreasing Co content as will be further discussed below. A contrasting response to pH variance is seen in the OCP values of AA 2024-T3 in the aerated 0.6 M NaCl solution as compared to the AA 2024-T3 and amorphous alloys in deaerated solution, with the OCP showing little variation as the pH of the test solution is raised. When tested in a pH 2 solution, the AA 2024-T3 has an OCP of –0.69 V_{SCE} and increases to –0.560 V_{SCE} in a solution of pH 12 owing to Cu replating.

The dissolution behavior for the amorphous Al-Co-Ce alloys was also shown to be significantly different from pure aluminum when tested as a function of pH (Figure 4). In 0.6 M NaCl solutions adjusted to acidic pH values, the amorphous alloys displayed a passive current density (i_{pass}) that was 2 to 3 times greater than that of the high purity aluminum samples. For example, the $AI_{87}Co_{8.7}Ce_{4.3}$ amorphous alloy had an i_{pass} of 5.4×10^{-5} A/cm^2 in a 0.6 M NaCl solution of pH 2, while the high purity Al had an i_{pass} of only 2.5×10^{-5} A/cm^2 in a similar solution. In contrast, the passive current density of the alloys was significantly lower than that of high purity aluminum in highly alkaline solutions. At a pH value of 11, the pure Al sample exhibited an i_{pass} of 1.25×10^{-4} A/cm^2 while the amorphous $Al_{87}Co_{8.7}Ce_{4.3}$ alloy had a value of only 1.6×10^{-5} A/cm^2. There were no detectable differences in the current densities of scans conducted over a range of near-neutral pH values. Current densities of the alloys heat treated to 100°C exhibited no significant deviation from the as-quenched amorphous material when similarly tested over the range of pH values, also seen in Figure 4.

Local Corrosion Properties

Alloys of the amorphous Al-Co-Ce system displayed pit formation but exhibited localized corrosion properties in deaerated 0.6 M NaCl solution that were significantly different from high purity polycrystalline aluminum, including a several hundred mV elevation of both E_{pit} and E_{rp}. Such significant differences are not a result of small at-risk areas or differing pit sizes, which were avoided by using similarly sized samples in all cases with pits grown to comparable depths[25, 10]. An example of this behavior is seen in Figure 5, where the fully amorphous $Al_{90}Co_7Ce_3$ sample has a pitting potential that is raised 328 mV above that of the high purity polycrystalline Al sample. In addition, the repassivation potential of the $Al_{90}Co_7Ce_3$ alloy is increased by 371 mV relative to the E_{rp} of pure Al. Such behavior was found in all of the fully amorphous alloys examined in the Al-Co-Ce system (3-11 at. % Co), as illustrated in Table 2, which summarizes the E_{pit} and E_{rp} results for a number of the compositions examined. For alloys shown in Table 2, the means of the E_{pit} values ranged from 450 mV to 487 mV higher than the mean E_{pit} value for pure Al, while the alloy's E_{rp} mean values ranged from 390 mV to 429 mV higher than that of the high purity Al. Also observed for the Al-Co-Ce system was the high statistical reproducibility of E_{rp} compared to the widely distributed values of E_{pit}, in agreement with results from other Al-based amorphous metals[9].

Tunable Corrosion Properties Based on Alloy Composition

Two of the selected local corrosion property metrics showed a direct dependence on alloy composition, specifically on the atomic percentage of cobalt. As cobalt content is varied from 0 to 10 atomic percent, the E_{rp} mean values for each composition follow a linear trend upwards from –0.85 V_{SCE} for alloys with 0% cobalt to span 400 mV, before reaching a plateau of –0.33 V_{SCE} at a cobalt value of 7% (Figure 6). This same response to cobalt composition was also seen in the mean OCP values, where compositions with 0% cobalt had a mean OCP value of –1.4 V_{SCE}, but as cobalt composition was increased to 7% the OCP rose to –0.6 V_{SCE} (Figure 7). It should be noted that Al-(Co)-Ce alloys containing 0-1% Co contained nanocrystals in the as-quenched condition as indicated by Figure 1. Similar influences were not found when E_{OCP} values were plotted as a function of the cerium or aluminum contents. However, when cobalt content was held constant at 7 atomic %, a slight decrease in the OCP was seen with increasing cerium content. This also held true for the E_{rp} values, suggesting that cobalt exerts the dominant influence on both properties, while cerium content plays a secondary role. When E_{pit} and i_{pass} behaviors were examined as a function of alloy composition, no statistically significant influences could be found for any of the three elemental components from potentiodynamic scans.

An additional investigation was carried out to determine if the Al-Co-Ce system retained its tunable OCP property in non-neutral solutions. A selection of alloys were tested in 0.6 M NaCl solution adjusted to either pH 3 or pH 10, and the OCP was measured after one hour exposure to the solution (Figure 7). The results indicate that, although there is a shift in the OCP values for each alloy composition as the solution pH changes, the OCP shows a similar dependence on cobalt concentration at each pH solution examined (i.e., the OCP increases with Co content).

Retention of Electrochemical Properties upon Low Temperature Aging

Scans conducted on Al-Co-Ce alloys artificially aged at 100°C showed excellent retention and selective improvement of the localized corrosion metrics (Figure 8). The mean values of both OCP and E_{rp} showed no significant variation as the amorphous material was heat-treated with increasing time duration at 100°C. However, the mean E_{pit} values of the 100°C heat treated samples were all elevated above that of the as-received material. Only those samples that were fully recrystallized (24 hours at 550°C) had a reduced mean pitting potential and lowered mean repassivation potential, as shown in Figure 9. Specific microstructural characterization through XRD is underway for each of these low temperature heat-treatments, allowing for the electrochemical behavior to be definitively linked to the development of nanocrystals.

DISCUSSION

All amorphous alloys examined in the Al-Co-Ce system were shown to have improved localized corrosion resistance as compared to high-purity aluminum, in the form of elevated E_{pit} and E_{rp} values (Figure 6). While the elevated E_{pit} values may be attributed to the amorphous microstructure of the alloy material[9], the altered repassivation potential is most likely due to the presence of TM in solid solution.

An additional effect of the TM and RE alloying elements is the altered pH dependent dissolution behavior which the amorphous alloys demonstrate when compared to the well-known behavior of pure aluminum (Figure 4). Pourbaix diagrams[26] for each of the alloying elements indicate that aluminum forms an insoluble hydroxide ($Al(OH)_3$) in the potential range investigated when pH ranges from approximately 3 to 10, while cobalt and cerium actively dissolve when pH values decrease below 6. Cobalt and cerium in substitutional solid solution in the amorphous phase are credited for the increased dissolution of the amorphous alloys when compared to high-purity aluminum examined in moderately acidic solutions. In contrast, when the amorphous alloys were examined in alkaline solutions, the dissolution rate of the alloys was much slower than high-purity aluminum due to likely formation of passivating insoluble oxides formed by both cobalt and cerium in this pH region (most likely $Ce(OH)_3$ and $Co(OH)_2$). The OCP values for amorphous (2-11% Co) and amorphous + nanocrystalline (0-1% Co) Al-Co-Ce alloys were also shown to be strongly influenced by the addition of alloying elements into the solid solution of aluminum, with cobalt raising the open circuit values significantly (Figure 8). These influences allow the OCP to be controlled through alloying additions, resulting in an amorphous alloy with tunable electrochemical properties.

Another parameter associated with localized corrosion that was found to be tunable based on alloy composition was E_{rp}. The increase of cobalt content in solid solution resulted in a dramatic elevation in the E_{rp} values (Figure 7). Such a strong influence is most likely due to either a decrease on the pit dissolution rate or the need for more concentrated critical depassivating pit solution for cobalt-containing alloys. Further investigations to isolate the mechanism by which cobalt elevates OCP and E_{rp} are currently underway.

CONCLUSIONS

1. Localized corrosion resistance in near neutral halide solution is enhanced for all fully amorphous alloys of the Al-Co-Ce system when compared to pure polycrystalline aluminum, in the form of elevated E_{pit} and E_{rp} mean values.
2. As cobalt concentration within the bulk amorphous alloy is raised, the open circuit potential and repassivation potential both show concurrent increases in their values when tested in near-neutral halide solutions.
3. Dissolution kinetics of the amorphous Al-Co-Ce alloys in acidic and alkaline solutions are altered from those of pure aluminum. The alloys show reduced dissolution in alkaline environments and increased dissolution in acidic environments.
4. Low-temperature heat treatments that result in structural relaxation and/or partial devitrification did not result in a loss of the enhanced corrosion properties shown by the as-quenched amorphous alloys. However, complete recrystallization negated all benefits of alloying and the amorphous state.

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Table I. Al-Co-Ce amorphous alloy compositions examined.

$Al_{93}Ce_7$	$Al_{92}Ce_8$	$Al_{92}Co_1Ce_7$	$Al_{91}Co_1Ce_8$	$Al_{90}Co_3Ce_7$	$Al_{89}Co_3Ce_8$
$Al_{84}Co_5Ce_{11}$	$Al_{90}Co_7Ce_3$	$Al_{87}Co_7Ce_6$	$Al_{86}Co_7Ce_7$	$Al_{83}Co_7Ce_{10}$	$Al_{87}Co_{8.7}Ce_{4.3}$

Table 2. Mean pitting and repassivation potentials for amorphous Al-Co-Ce alloys and high purity polycrystalline aluminum (99.999%) with ~10^{-4} cm^2 surface areas tested in deaerated 0.6 M NaCl solution (pH 6).

	E_{pit} (V_{SCE})	E_{rp} (V_{SCE})
Pure Aluminum (polycrystalline)	-0.701 ± 0.016	-0.736 ± 0.001
$Al_{90}Co_7Ce_3$ (amorphous)	-0.214 ± 0.020	-0.307 ± 0.008
$Al_{87}Co_{8.7}Ce_{4.3}$ (amorphous)	-0.218 ± 0.028	-0.346 ± 0.017
$Al_{87}Co_7Ce_6$ (amorphous)	-0.252 ± 0.026	-0.317 ± 0.016
$Al_{87}Co_7Ce_6$ (fully crystalline)	-0.674 ± 0.055	-0.742 ± 0.015

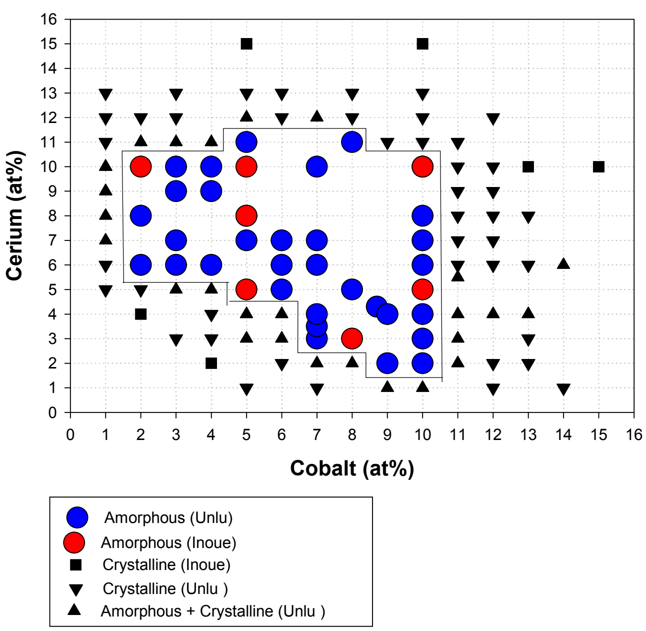


Figure 1. Al-Co-Ce alloy system map designating amorphous compositions developed by Unlu and Inoue[21] for alloys produced with 52 m/s circumferential wheel speed, and with microstructure determined through X-ray diffraction, TEM bright field analysis, and TEM diffraction.

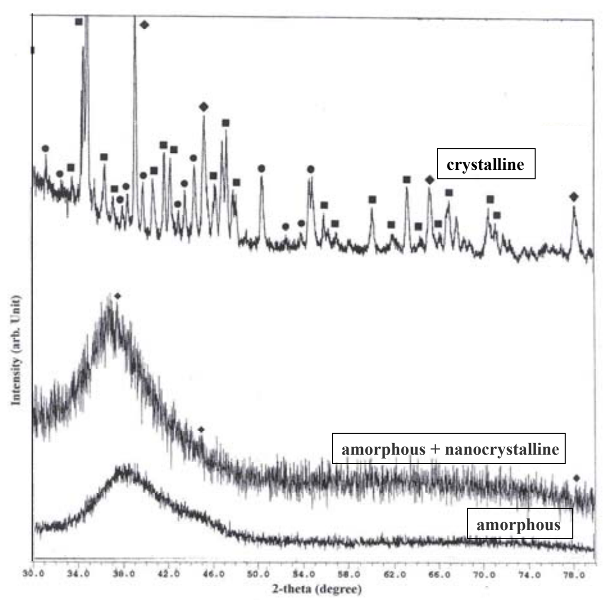
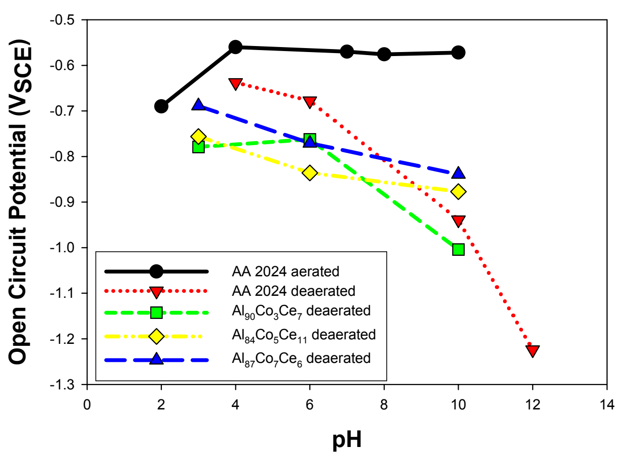


Figure 2. X-ray diffraction patterns (Cu K_{α} radiation at 40 kV and 35 mA) for the melt-spun $Al_{87}Co_{8.7}Ce_{4.3}$ alloy ribbon (52 m/s wheel speed) showing the amorphous microstructure of the as-quenched material, the amorphous + nanocrystalline microstructure of the 100 hour 100°C treatment, and the crystalline microstructure which is developed after a non-isothermal 550°C heat-treatment.



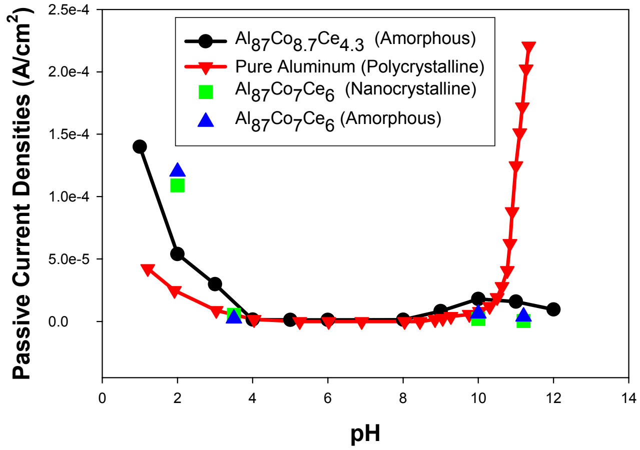
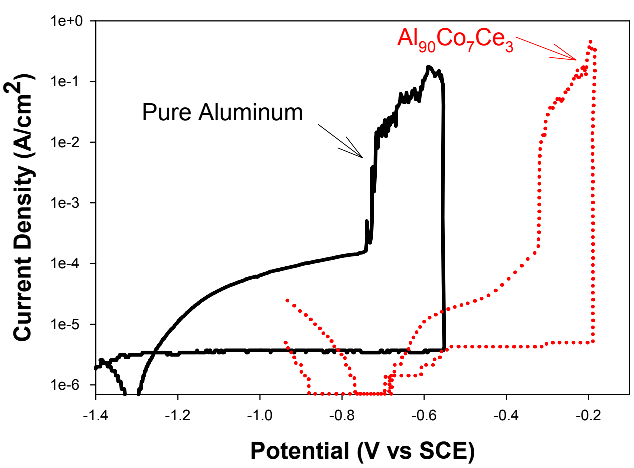


Figure 3. Open circuit potentials for amorphous Al-Co-Ce alloy and AA 2024 after one-hour exposure to 0.6 M NaCl solution with pH adjusted using 1 M HCl or 1 M NaOH as necessary.

Figure 4. Passive current densities for high purity Al[27] and amorphous alloys of the Al-Co-Ce system, including heat-treated nanocrystalline material, in solution adjusted to various pH values using concentrated HCl or 1 M NaOH as neccesary.



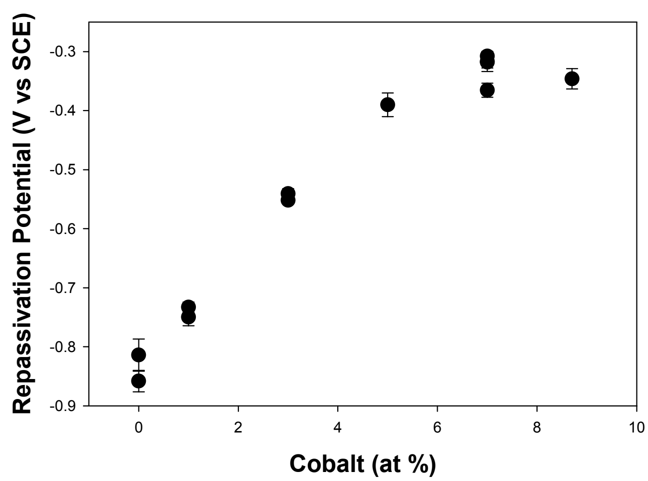


Figure 5. Potentiodynamic scans conducted on amorphous Al-Co-Ce alloy and 99.999% aluminum in deaerated 0.6 M NaCl solution (pH 6). Both working electrode areas were ~10^{-4} cm^2.

Figure 6. Repassivation potential for as-spun Al-Co-Ce alloys with 0 to 8.7 atomic percent cobalt in deaerated 0.6 M NaCl solution (pH 6) expressed as a function of cobalt. Alloys with >1% Co are fully amorphous while those with 0-1% Co contain nanocrystals.

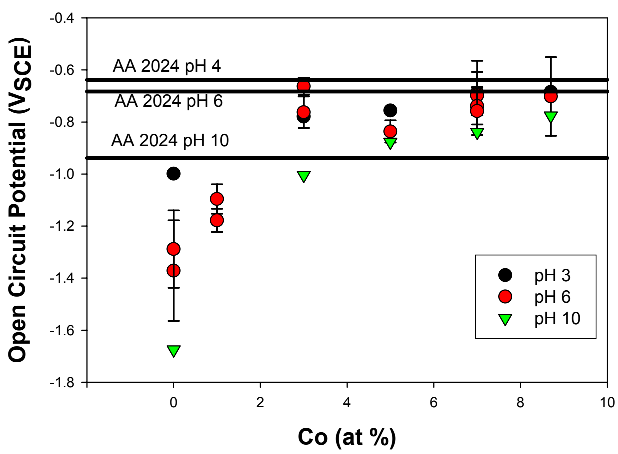


Figure 7. Open circuit potentials of as-spun Al-Co-Ce samples after one-hour exposure to deaerated 0.6 M NaCl solution (pH 6) expressed as a function of atomic percent cobalt. Alloys with greater than 1% Co are fully amorphous. Data compared to AA 2024-T3 in deaerated 0.6 M NaCl selected pH solutions.

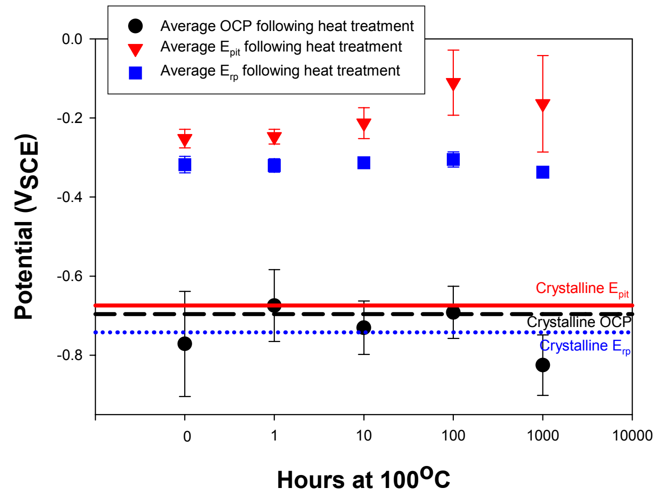


Figure 8. Effect of 100°C isothermal heat treatments on E_{pit}, E_{rp}, and OCP values for $Al_{87}Co_7Ce_6$ alloy when tested in deaerated 0.6 M NaCl after a one hour exposure to solution. Data is compared to fully crystalline $Al_{87}Co_7Ce_6$ alloy after 550°C isothermal heat treatment.

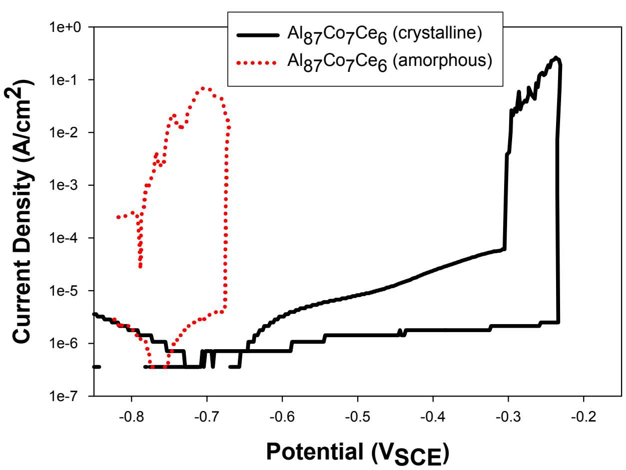


Figure 9. Potentiodynamic scans conducted on amorphous and fully crystalline (following 550°C isothermal heat treatment) $Al_{87}Co_7Ce_6$ alloys in deaerated 0.6 M NaCl solution.